

Final Tech Stack: Full List with APIs

AI-Based Compliance Monitoring for Skill Training Centres | SIH 2026, PS 26245

Everything is from the project PDF. Items marked (*proposed*) were added by us and are not in the PDF.

1. Trainee App (on the phone)

Technology	What it does	Why we use it
Flutter	Single app for trainees and trainers	One codebase, good camera and Bluetooth support
TFLite (INT8)	Runs compressed AI models on the phone	Works offline, runs on cheap and old phones
MobileFaceNet / ArcFace	Turns the live face into numbers and matches it with the enrolled face	Stops "phone handed to a friend"
ML Kit / MediaPipe	Live challenge (blink twice, turn left)	Works offline on the phone
MiniFASNet	Detects photo, screen or deepfake faces	Small enough for a phone
BLE peer scanning	Phones in the room detect each other	A phone no one else saw is probably not in the room
Play Integrity API	Detects rooted phones, emulators, tampered apps	Hardware-backed, hard to forge
Android Keystore	Signs every check-in with a hardware-protected key	Packets cannot be forged or reused

2. Room Beacon (rotating code)

Technology	What it does	Why we use it
ESP32 board (~₹400)	Fixed on the classroom wall, broadcasts the room code	Very cheap, low power, easy to deploy
TOTP (6-digit code, rotates every 30 s)	Beacon and server compute the same code from a shared secret and the current time	Forwarded or screenshot codes expire in 30 s
BLE broadcast	Sends the current code, range about 10 m	Phones already have Bluetooth. Marking from home fails
Sound token (optional)	Beacon also plays the code as a soft sound the phone mic must hear	Sound does not pass walls easily, so it helps against someone standing outside

3. Edge Unit (inside CCTV centres)

Technology	What it does	Why we use it
Mini PC / Jetson-class device	Runs the AI inside the centre	Video stays local, saves bandwidth, protects privacy
ONNX / OpenVINO / TensorRT	Speeds up AI models on small hardware	Real-time on low-cost devices
CLAHE	Fixes dark camera frames	Helps in low-light classrooms
YOLO (v8/11) / RT-DETR	Counts people, benches, machines, chairs, computers	Fast and accurate on small devices
ByteTrack	Follows each person across frames	Nobody is counted twice
InsightFace (SCRFD / RetinaFace)	Finds faces in CCTV frames	Strong open-source models
ArcFace / AdaFace	Matches faces against this batch only (~30 people)	Fewer candidates means fewer wrong matches

4. Sensors (for equipment)

Technology	What it does	Why we use it
PZEM-004T / smart plug (₹500 to 800 per machine)	Measures current drawn by a machine	A lathe "used for 6 hours" with no current is a clear flag
Vibration sensor / MEMS mic	Detects whether the machine is running	Cheap and objective proof of use

5. Communication

Technology	What it does	Why we use it
MQTT (Mosquitto)	Sends about 1 KB signed events from centres	Built for weak networks, supports offline
HTTPS	Sends events from phones	Standard and secure
Offline queue + signed timestamps	Phones and Edge Unit store events and sync later	Works with no internet, cannot be backdated

6. Backend and Storage

Technology	What it does	Why we use it
FastAPI	Our backend and REST APIs	Fast, simple, scales well
PostgreSQL (PostGIS)	Main database, centre locations	Reliable. A unique rule on (student, session, prompt) ignores duplicates
Redis + Celery	Queue for bursts, scheduled random check-in prompts	30 students tapping at once will not lose data
MinIO / S3	Stores blurred flag thumbnails (about 15 to 20 KB)	Low cost, access controlled

7. Decision Engine (AI/ML)

Technology	What it does	Why we use it
Rules + two-signal rule	One weak signal never raises a flag	Fewer false accusations
Logistic regression / XGBoost (scikit-learn)	Learned, calibrated scoring	"0.9" really means about 90%
Isolation Forest	Finds odd patterns over time	Catches things like a centre at 100% every day
MLflow	Tracks model experiments	Easy to audit

8. Monitoring Dashboard

Technology	What it does	Why we use it
React	Officer dashboard	Fast to build clear views
Leaflet	Map of centres coloured by compliance	Problem areas stand out
Recharts	Claimed vs detected charts, trust distribution	Simple charts
Streamlit	Dashboard for the prototype only	Quickest way to demo

9. Security and Deployment

Technology	What it does	Why we use it
TLS	Encrypts data in transit	Standard security
AES-256	Encrypts stored data and face embeddings	Strong protection at rest
JWT + role-based access	Ministry, state, district and centre see different views	Controls who sees what
Audit logs	Records who viewed what	Accountability
Docker Compose	Packages the system	Same setup on any server

APIs and Interfaces

API / Interface	Used for	Status
Google Play Integrity API	Confirms the phone and app are genuine	In PDF
Android Keystore	Hardware-backed signing (Android system feature)	In PDF
Our FastAPI REST APIs	Check-in submit, alerts, reports, dashboard data	In PDF
MQTT / HTTPS ingestion	Edge Unit and phone events	In PDF
Aadhaar eKYC (where permitted)	Identity check at enrolment	In PDF, needs legal approval. Say "scheme ID verification, Aadhaar eKYC where permitted"
Push notifications	Random-time check-in prompt	PDF says the server sends it but names no service. Firebase Cloud Messaging is the usual choice (<i>proposed</i>)
Government / scheme reporting APIs	Link with scheme systems	Planned for the National phase only, not yet validated

Short version for your slide (one table)

Part	Tech	Why
App	Flutter, TFLite	One codebase, offline AI
Face + liveness	MobileFaceNet/ArcFace, MiniFASNet, MediaPipe	Stops proxy and fake faces
Phone trust	Play Integrity API, Keystore	Blocks fake phones
Room proof	ESP32 beacon, BLE, TOTP (6-digit code, rotates every 30 s), optional sound token	Cheap; forwarded codes expire
Camera AI	YOLO, ByteTrack, InsightFace	Count and match
Edge	Mini PC/Jetson, ONNX	Video stays local
Sensors	PZEM-004T, vibration	Proof a machine ran
Messaging	MQTT	Weak network, offline
Backend	FastAPI, PostgreSQL, Redis/Celery	Scalable
Decision	scikit-learn, XGBoost, Isolation Forest	Explainable
Dashboard	React, Leaflet, Recharts	Map and alerts
Security	TLS, AES-256, JWT, audit logs	Role-based access

Judge-ready notes

- The only third-party API we rely on by name is **Play Integrity**. Do not claim live government API integration.
- **Android-first** is a pilot scope decision, because Bluetooth peer scanning and Play Integrity work best on Android.
- Be ready to answer: "What if beacon and server clocks drift?" A safe answer is a small time allowance for the previous or next 30 s code (*proposed, not in the PDF*).